Pranav Senthilkumar

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Education

Texas A&M University College Station, TX
Bachelor of Science in Electrical Engineering, Engineering Honors *Expected May 2028*
- **Coursework:** Digital Logic Design, Machine Learning, Multivariable Calculus, Differential Equations, Data Structures & Algorithms

Technical Skills

Languages: Embedded C++, Python, Verilog, Java, JavaScript, SQL
Embedded: ESP32, Raspberry Pi, Arduino, Linux, ROS 2
Hardware: Digital Logic Design, Computer Architecture, KiCad,RTL
ML & Vision: PyTorch, OpenCV, ONNX Runtime, scikit-learn, MediaPipe
Tools: Linux, Git, GitHub Actions, Docker, AWS, Electron

Experience

Consultancy Team Member September 2026 – Present
Aggies Create, Shallow Sports Boats *College Station, TX*
- Implementing C++ firmware on an ESP32 to stream GPS, IMU, sonar, and propulsion telemetry to the navigation stack
- Deploying a quantized vision model on a Raspberry Pi for real-time water segmentation and obstacle detection
- Designing C++ and ROS 2 navigation software that turns mission-level objectives into autonomous survey behavior

Robotics Software Engineering Intern August 2026 – September 2026
Nietzsche Labs *Remote*
- Shipped a cross-platform Electron app that provisions robot OS images onto SD cards, backed by an AWS serverless API
- Enforced **8 pre-write safety checks** with target-drive re-verification to prevent accidental overwrites of host disks
- Automated first-boot partition expansion on embedded Linux images, reclaiming **90–97%** of storage stranded behind a fixed 2.75 GiB partition, validated on 10 card sizes (2–256 GB) with zero failures

Undergraduate Machine Learning Researcher June 2026 – Present
LIVE Lab, Texas A&M University *College Station, TX*
- Building ML models to turn regional wind forecasts into street-level predictions across urban and open terrain
- Developed an end-to-end urban wind pipeline combining LiDAR, terrain, and building data with a 3D model, cutting near-ground error by **50%** and prediction time from ∼3 min to under 1 s
- Improved 4× wind-field resolution by **59.9%** over interpolation using a residual correction model

Student Researcher June 2024 – October 2024
Algoverse AI Research *Remote*
- Calibrated large language models to reflect the spread of human opinion on morally ambiguous questions
- Fine-tuned 3 open-source LLMs with parameter-efficient training, cutting error versus human responses by up to **31%**
- Matched **GPT-4o** on moral reasoning using a 7B open-source model, resulting in a NeurIPS 2024 workshop paper

Projects

CrossLayer | *RTL, Verilog, RISC-V, Python, Icarus Verilog* October 2026 – Present
- Built a verification framework with pluggable adapters and behavioral contracts for validating RTL and software execution
- Integrated PicoRV32 with Icarus Verilog and an RV32I model, pinpointing 6 injected RTL defects
- Optimized event analysis by 32×, reducing processing time for 80,000 events from 6.6s to 0.21s

Regdrift | *Python, CMSIS-SVD, pytest, GitHub Actions* June 2026 – July 2026
- Created a CI tool that blocks **22 classes** of breaking register-map changes before they surface as firmware faults
- Engineered deterministic comparisons of registers, bit fields, interrupts, and peripherals for reproducible reports
- Validated with **330 automated tests** across 15 microcontroller definitions from 5 semiconductor vendors

Haptic Portal | *Python, DepthAI, OpenCV, MediaPipe* January 2026 – May 2026
- Prototyped a system that turns camera hand and depth tracking into vibration feedback, winning **Best Pitch** and $250
- Downsampled depth frames into a 25-zone (5×5) actuation grid, sustaining **30 Hz** feedback updates for responsive touch
- Applied temporal and median filtering to suppress depth-sensor noise before driving a custom motor array for stable feedback

ThermGuard | *Python, PyTorch, Conformal Prediction* May 2026 – June 2026
- Architected a thermal-aware scheduler that assigns workloads to cores using predicted per-core temperature limits
- Trained calibrated models whose temperature limits held in **95%** of cases against a 90% safety target across 5 trials
- Stress-tested reliability with an automated evaluation pipeline that detects failures under shifting workloads

Leadership

Treasurer & Architecture Team Member August 2026 – Present
Rev Silicon *College Station, TX*
- Managing funding and growth pathways for a student silicon design organization pursuing its first chip fabrication

Publication

Pranav Senthilkumar, et al. "Fine-Tuning Language Models for Ethical Ambiguity: A Comparative Study of Alignment with Human Responses." NeurIPS 2024 SoLaR Workshop | arXiv:2410.07826